

Introduction

Every year **600 billion dollars** are spent on manufactured plastics. The primary reason 3d printing isnt a major player in the idustury is that, parts made from many 3d printers aren't strong enough for many applications like limited run custom components and rapid prototyping. This project begins to bridge strength gap between **3d printed parts** and **injection molded parts**, while maintaining the cost benefits of avoiding injection molding by using a combination of inexpensive **chemicals** and a process simmlar to **vapor smoothing**. Increasing the strength of printed components. Resulting in the availability of **strong and affordable** custom parts for a variety of applications, and billions of dollars in opportunities.

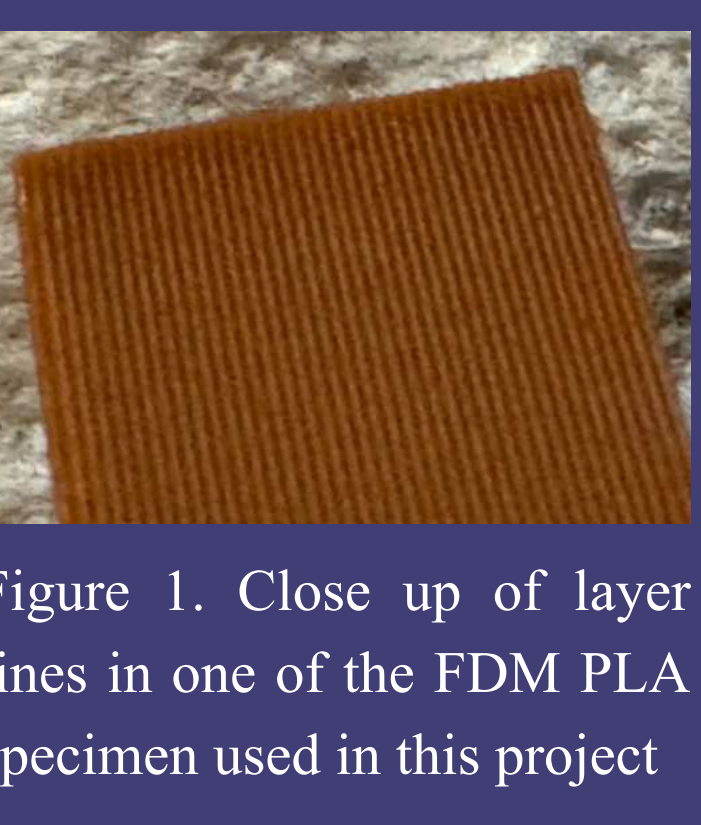


Figure 1. Close up of layer lines in one of the FDM PLA specimen used in this project

Fused Deposition Modeling (FDM) printers lack sufficient strength and durability for long term and high stress applications. Parts are **10-20%** below the minimun strength threshold for being suitably resistant to wear, cracking, and failure under repeated use or heavy loads. One potential solution involves coating 3D printed parts with acid or specialized adhesives to **enhance layer adhesion**, wear resistance, and structural integrity. Without such reinforcement, prints often underperform in real world environments, limiting **innovation** and **efficiency** in prototyping.

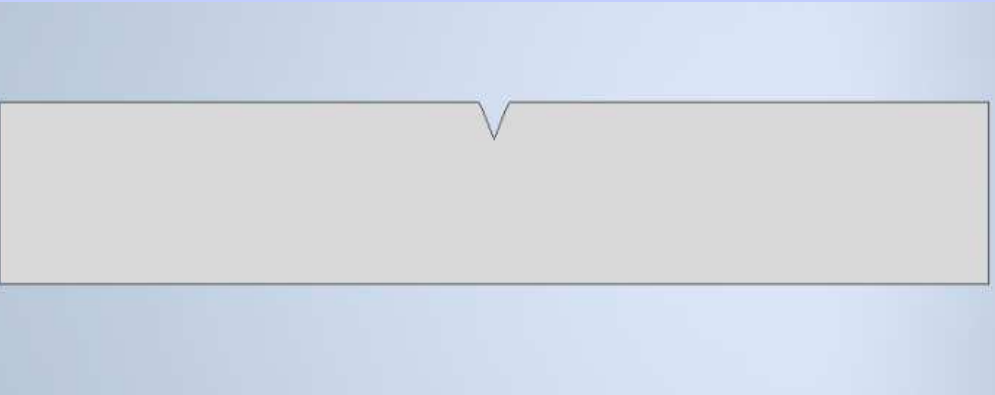


Figure 2. Charp impact specimen, This is the model used in this project, it was chosen because it was suitable for the testing equipment available and easy to print in large quantities.

Purpose

The aim of this research is to create a chemical formula that when applied to PLA 3d prints, increases and **impact** strength. The chemicals were chosen for their **low cost** and **non-toxicity**. The formula will help small businesses create **limited run** parts before committing the hundreds of thousands dollars required for access to **molded plastics** and allow for consistent rapid prototyping of complex models.

The formula will help improve the **adhesion** between layer lines by melting the outermost layer of material. As the material solidifies the theory is that the layers will blend with the surrounding layers. The improvement in bond strength is intended to elevate the **consistency** and minimum **strength** expected from 3d printed parts.

Optimizing Layer Adhesion Strength for Fused Deposition Modeling using Acetone and Cyanoacrylate

Methods

Initial Testing

Figure 3. This shows the Charpy Impact specimen that is used in this project, it was modeled to the standard specifications using CAD and 3d-printed using the Bamboo Labs X1 Carbon.

Figure 4. This test was used to collect data for the baseline impact strength of the 3d printed PLA specimens. This data will be used later to prove the effectiveness of the chemical coating. The impact test was chosen because it is representative of real world forces.

Figure 5. Initial testing showed that pieces made from red PLA were stronger in peak absorption, but brown PLA pieces were more consistently strong. The IQR of the brown piece was that of the red pieces, because of the increased consistency of the brown prints they were chosen for the remainder of the project.



Figure 3 - PLA Specimen

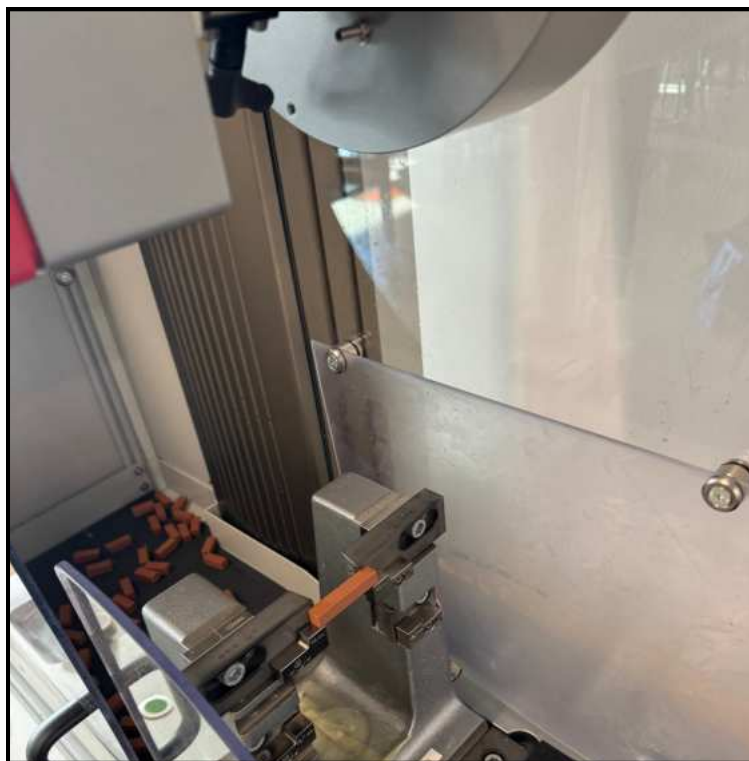


Figure 4 - Impact testing

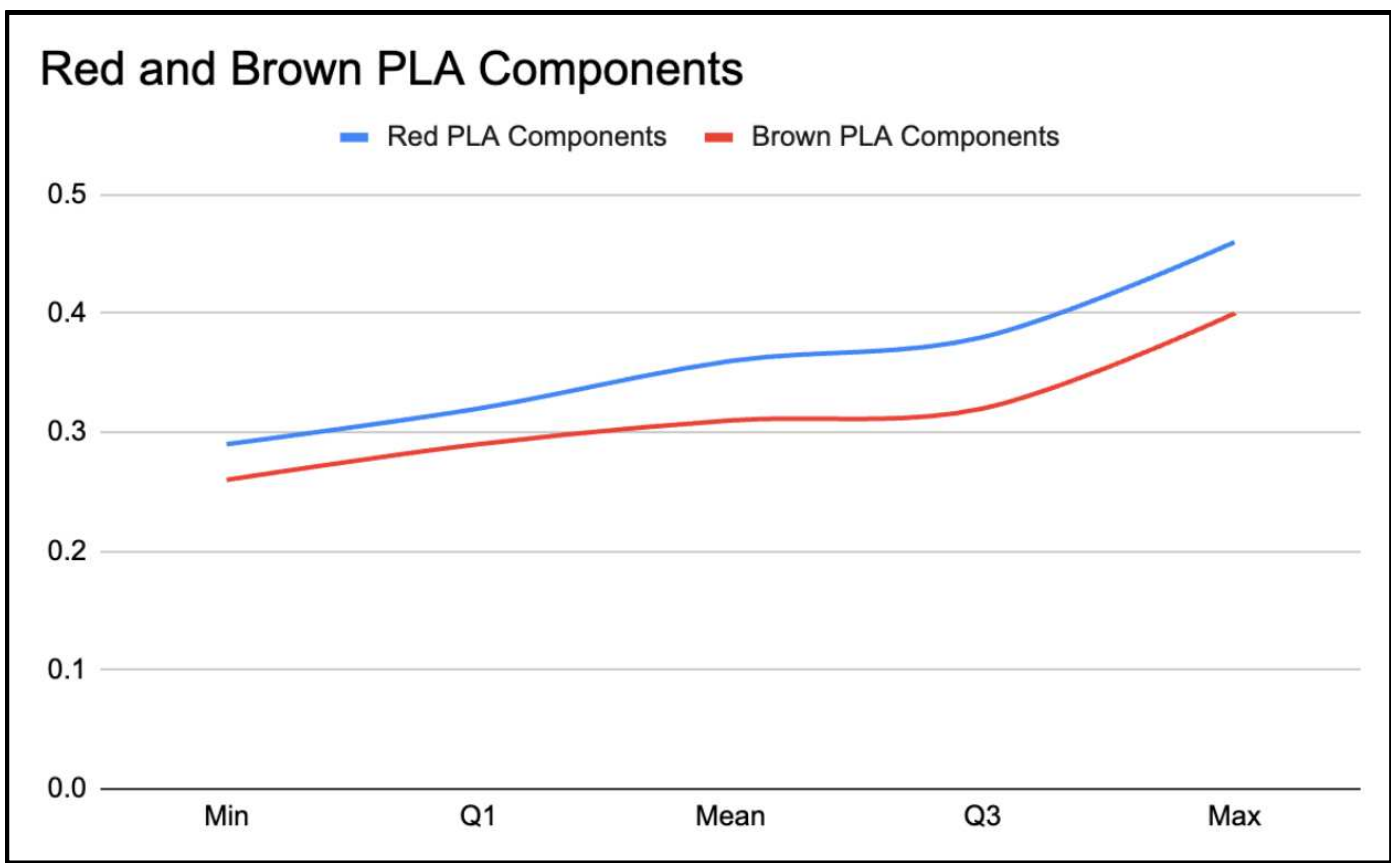


Figure 5 - Red and Brown impact testing comparison

Chemical Combination



Figure 6- The 3 glass jars shown in each figure are 1000ml, they were filled with 350 ,300, 250ml of water respectively. Then filled to 400ml with acetone. The solution was then mixed together with a glass stirring rod.



Figure 7- Shows the post chemical application area, on the white paper towel are the glass jars shown in the previous figure. The bottom of the figure shows the bags filled with each group's specimen with the notecards above them indicating what group and compound they are.



Figure 8 - These bags were labeled with group and exposure information then, tested indavidually in class.



Figure 9 - Shows the final result of coating, the strength/exposure increases from left to right.

Impact Testing

Featured below are the results from the impact testing of the coated parts, The most representative measurement for this project is EcJ, which is a measure of the force that was used to break the specimen. Group 1, 2, and 3 correspond to the strength of chemicals used on the part, while exposure time was five or ten minutes and was used to see if there was a point of diminishing return or even if too much exposure damaged the specimen more than it strengthened them.

Specimen no.	Angle of rise °	Ec J	Ec(%) %	Ec(Friction) J	Impact energy/Notch length J/m	a _{cN} kJ/m ²
11	109.5	0.34	6.3	0.01	6.17	0.77
6	56.18	0.03	0.6	0.00	0.57	0.07
56.34	-	9.20	9.20	0.31	9.20	9.20

Figure 10. - Group 1, 5 minute exposure.

Specimen no.	Angle of rise °	Ec J	Ec(%) %	Ec(Friction) J	Impact energy/Notch length J/m	a _{cN} kJ/m ²
12	130.8	0.34	6.2	0.01	6.10	0.76
6	29.21	0.02	0.4	0.00	0.44	0.06
56.47	-	7.24	7.24	0.24	7.24	7.24

Figure 12. - group 1, 10 minute exposure.

Specimen no.	Angle of rise °	Ec J	Ec(%) %	Ec(Friction) J	Impact energy/Notch length J/m	a _{cN} kJ/m ²
11	116.5	0.33	6.2	0.01	6.05	0.76
6	50.23	0.04	0.8	0.00	0.76	0.10
56.34	-	12.63	12.63	0.42	12.63	12.63

Figure 14. - Group 2, 5 minute exposure.

Specimen no.	Angle of rise °	Ec J	Ec(%) %	Ec(Friction) J	Impact energy/Notch length J/m	a _{cN} kJ/m ²
10	98.87	0.35	6.4	0.01	6.31	0.79
5	63.1	0.04	0.7	0.00	0.69	0.09
56.20	-	10.88	10.88	0.38	10.88	10.88

Figure 11. - group 2, 10 minute exposure.

Specimen no.	Angle of rise °	Ec J	Ec(%) %	Ec(Friction) J	Impact energy/Notch length J/m	a _{cN} kJ/m ²
12	124.3	0.35	6.4	0.01	6.30	0.79
6	40.25	0.04	0.7	0.00	0.68	0.09
56.47	-	10.85	10.85	0.37	10.85	10.85

Figure 13. - group 3, 5 minute exposure.

Specimen no.	Angle of rise °	Ec J	Ec(%) %	Ec(Friction) J	Impact energy/Notch length J/m	a _{cN} kJ/m ²
9	104.2	0.35	6.6	0.01	6.44	0.80
5	59.57	0.04	0.7	0.00	0.68	0.09
56.11	-	10.63	10.63	0.37	10.63	10.63

Figure 15. - Group 3, 10 minute exposure.

Analysis

These graphs are representative of the strength and consistency of the different groups of components. The initial uncoated parts had an IQR of .03J and a mean of .31J across the sample. Nearly all of the coated parts were stronger than the uncoated parts in impact tests.

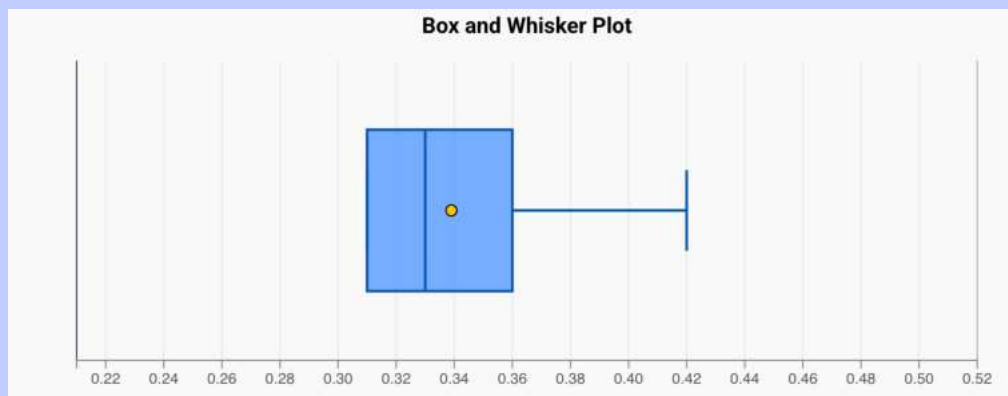


Figure 16. Group 1, 5 minute exposure - This was the worst performing group it had the weakest mean and a very large range of .09J, showing the compound was not given enough time to effective coat the parts

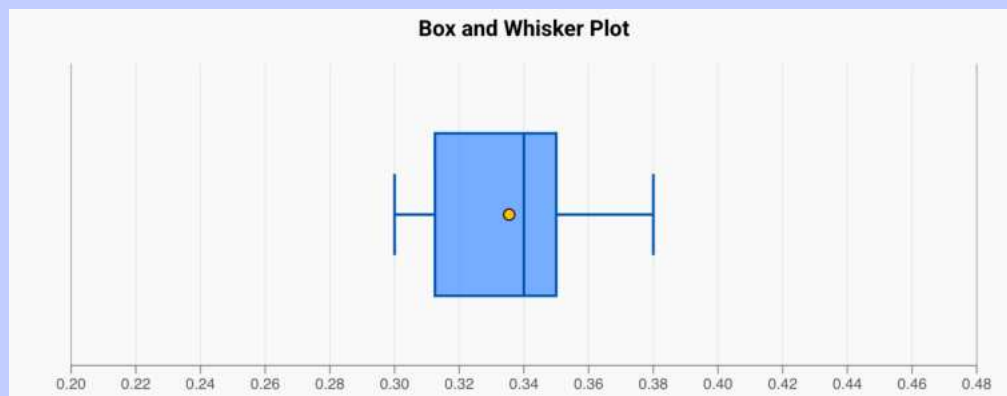


Figure 17. Group 1.10 - Longer exposure should have improved the minimum strength of the components, as the chemical would have more time to work on the most damaged sections, this was not supported by the results of the group.

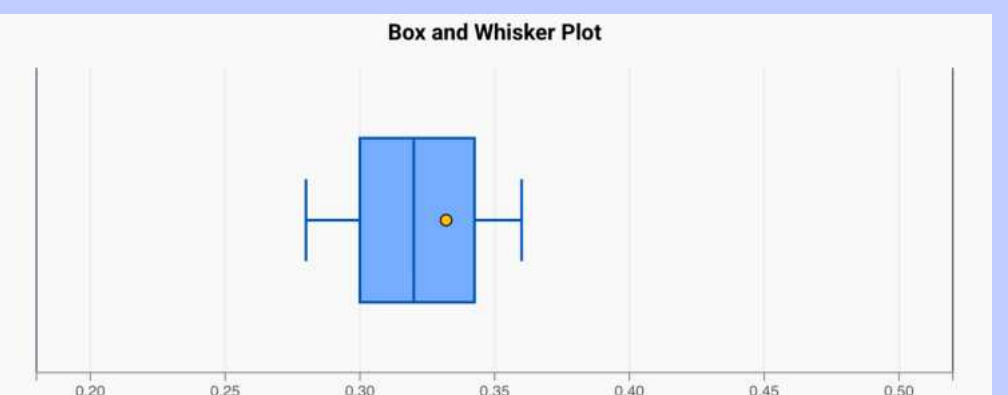


Figure 18. Group 2, 5 minute exposure - This group was the worst performing in the entire test, There was no meaningful improvement compared to the uncoated parts.

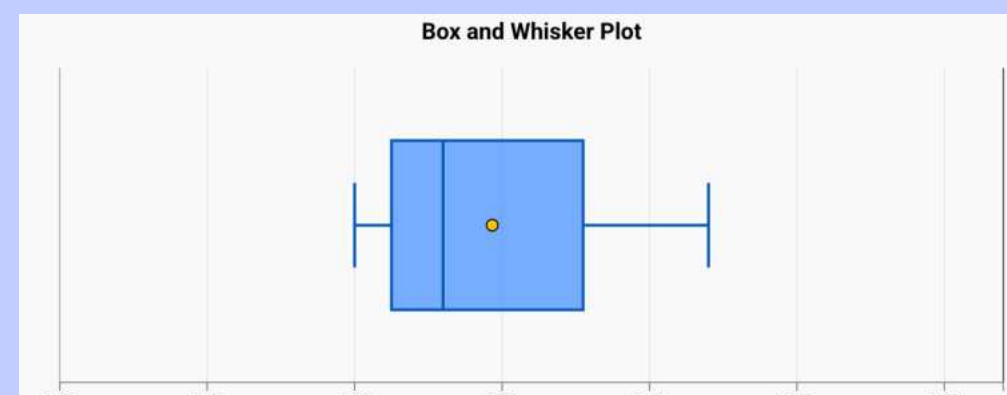


Figure 19. Group 2, 10 minute exposure - The extra 5 minutes of exposure for group 2 improved the results of some specimens. The strength of the group was .06J better than the five minute exposure.

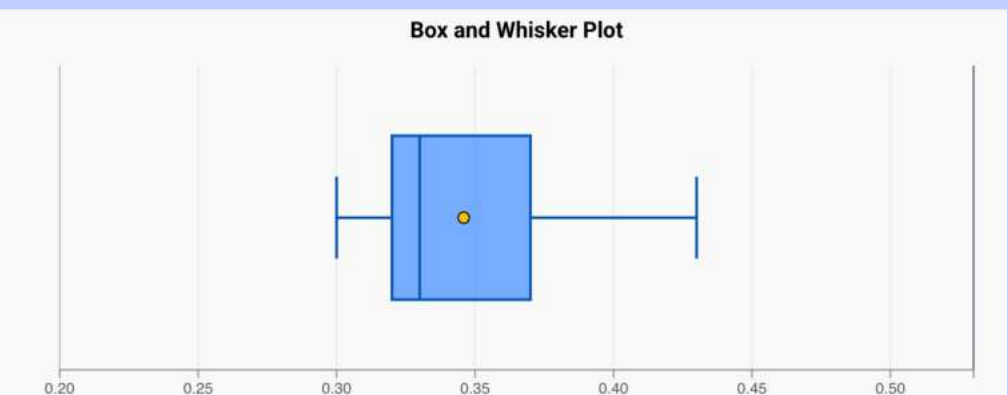


Figure 20. Group 3, 5 minute exposure - One difference in the data is between the 3.05 and 3.10 groups. Both the groups were the top of the test, but it seems that the longer exposure in the 10 minute group was beneficial.

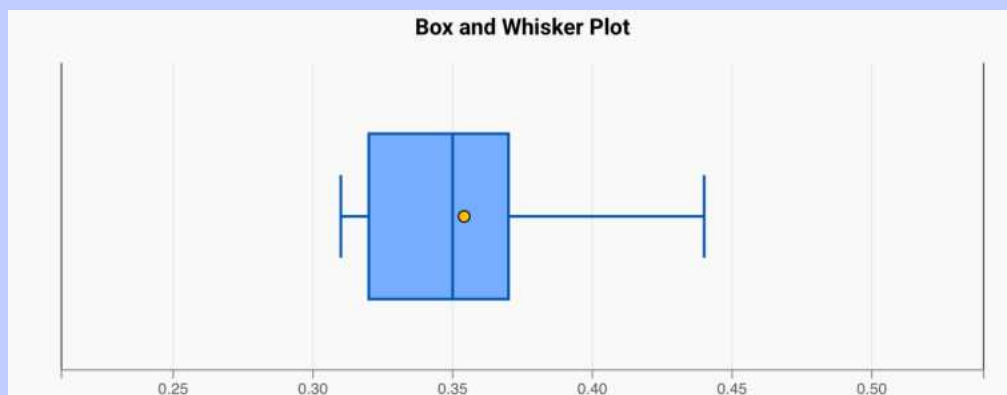


Figure 21. Group 3, 10 minute exposure - This is the most effective formula, the mean was .35J, but the more important metric was the IQR was .32J-.37J. This demonstrates improvement in every metric over the uncoated part.

Conclusion

The data collected in this project showed promise for the expanded use of 3d printing, the **most effective concentration** brought a **~7%** increase in mean strength and a **~15%** increase in minimum strength. In this test the most effective formula was 150ML of acetone, 250ML of water, and 1 gram of Cyanoacrylate. Mean strength went from **.31J** to **.33J**, minimum strength improved from **.26J** to **.30J**, and max strength improved from **.40J** to **.43J**. This improvement is statistically **significant** at the .05 level. Further applications for this project includes narrowing in on the most optimal ratio of acetone, as the best concentration is likely between **100-150ML** per 400 ML solution. Evaluating more precise exposure times could also yield improvements over what was seen in this experiment.

References

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